

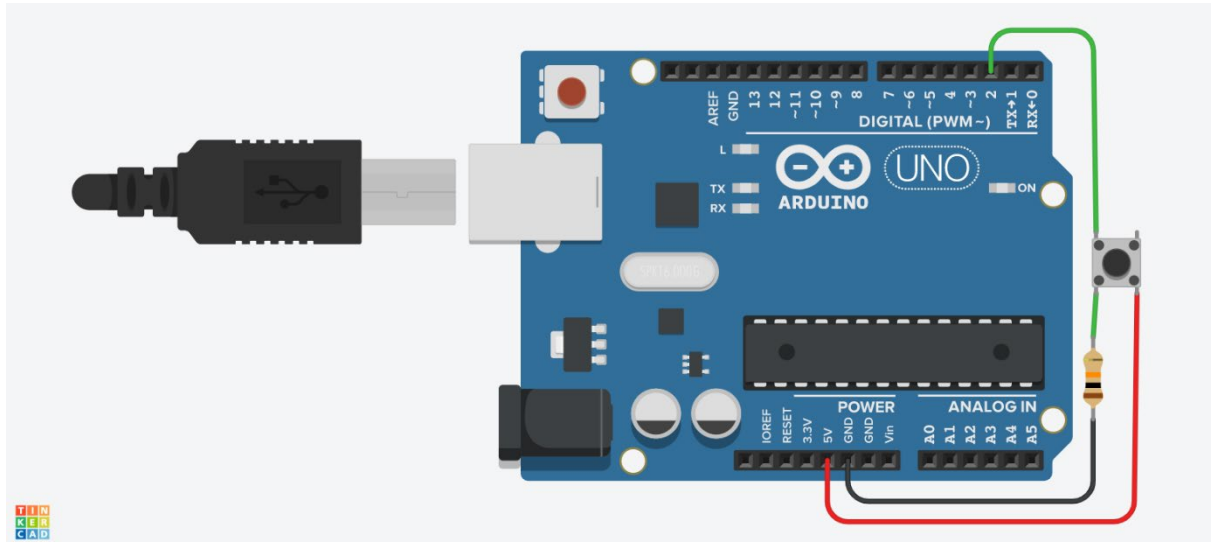
Input Operations & Introduction to Buttons

TinkerCAD Activity: "Input Operations & Introduction to Buttons"

1. "1. Introduction to working with Inputs"

- a. Go to "Activities", "Copy and Tinker" the associated task, and update the file name.
- b. Construct the following:
 - i. Arduino Uno R3
 - ii. 1 Pushbutton
 - iii. 1 10K resistor
 - iv. Built-In LED on pin 13.

Make sure the pins are connected exactly as you can see on the following image:

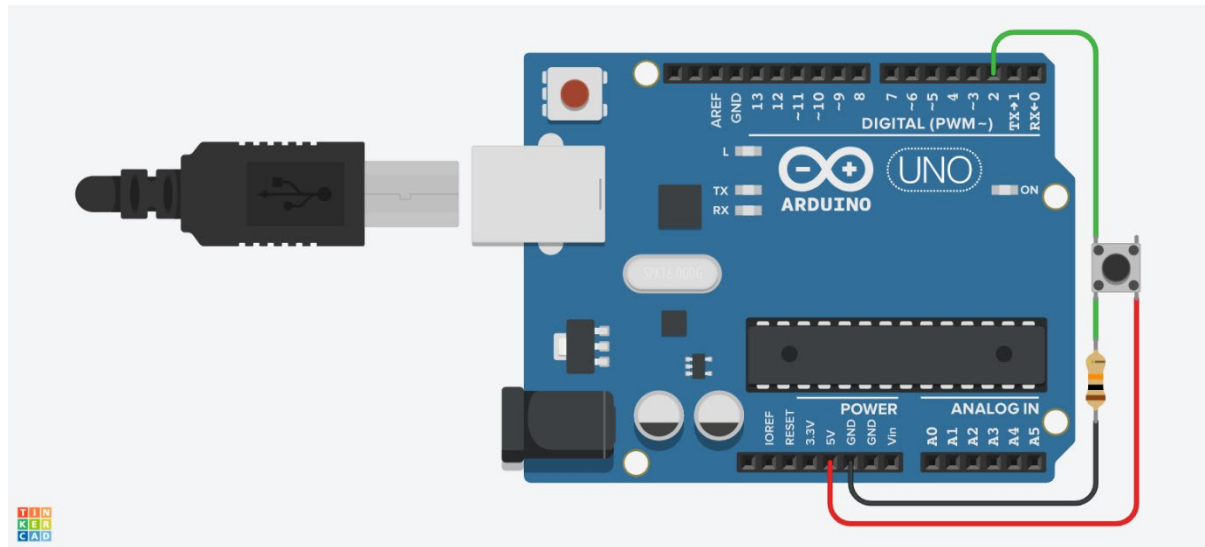


- c. Write a code that does the following:
 - i. Setting relevant pins as INPUT/OUTPUT.
 - ii. When the BUTTON is PRESSED - TURN-ON the Built-In LED (pin 13).
When the BUTTON is RELEASED - TURN-OFF the Built-In LED.

2. “2. Debounce Simulation”

- a. Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- b. Construct the following structure:
 - i. Arduino Uno r3.
 - ii. 1 Pushbutton.
 - iii. 1 10Kohm resistor.
 - iv. Built-In LED connected to pin 13.

Make sure the pins are connected exactly as you can see on the following image:



- c. Write a code that does the following:
 - i. Setting relevant pins as INPUT/OUTPUT.
 - ii. Handle the “Debouncing” using a 50ms delay (programmatically).
This means a “debouncing functionality” should be added to handle every button PRESS/RELEASE.
 - iii. Coordinate the BUTTON PRESS to TURN-ON the Built-In LED (pin 13) and BUTTON RELEASE to TURN-OFF the Built-In LED.

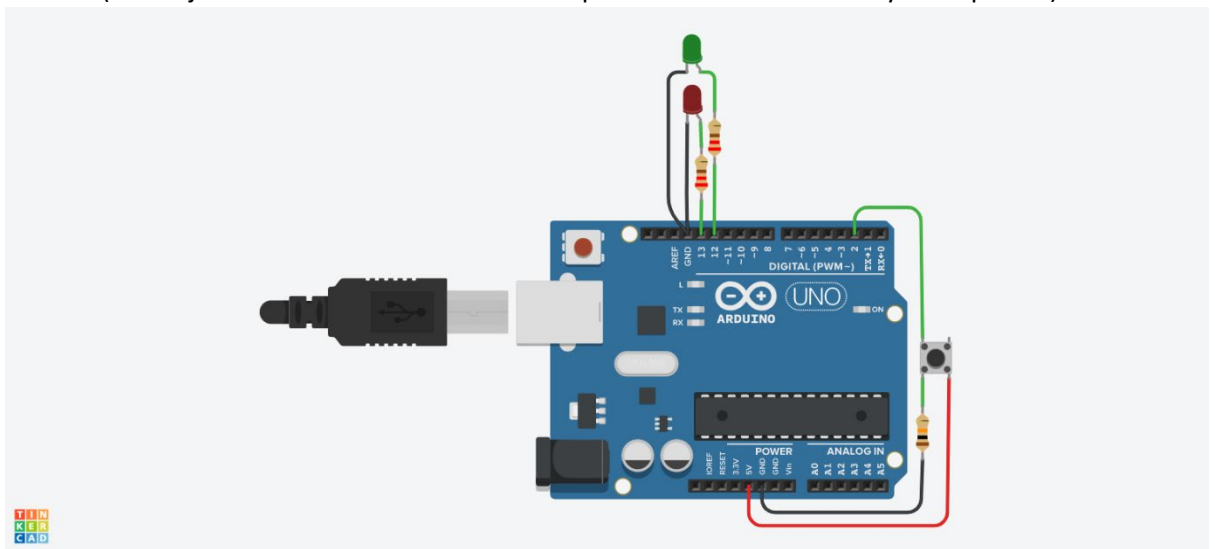
NOTES:

The changes/additions from the previous exercise might very subtle. It's OK! (just adding debouncing handling) 😊

3. “3. LED Switcher”

- a. Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- b. Construct the following structure:
 - i. Arduino Uno r3.
 - ii. 1 Pushbutton.
 - iii. 1 10Kohm resistor.
 - iv. 1 Red LED on pin 2
 - v. 1 Green LED on pin 5
 - vi. 2 220ohm resistors.

(This is just an EXAMPLE. Make sure the pins are connected exactly as required!)

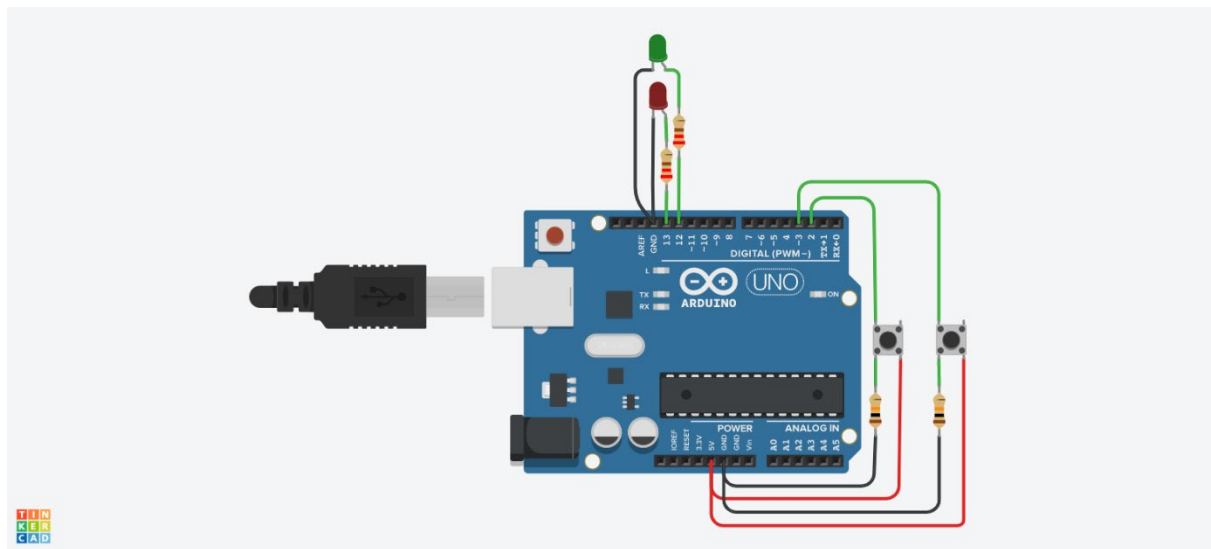


- c. Write a code that does the following:
 - i. On button press → Green ON, Red OFF
 - ii. On button release → Green OFF, Red ON

4. "4. 2 Buttons + 2 LEDs"

- a. Go to "Activities", "Copy and Tinker" the associated task, and update the file name.
- b. Construct the following structure:
 - i. Arduino Uno r3.
 - ii. 2 Pushbuttons.
 - iii. 2 10Kohm resistor.
 - iv. 1 Red LED on pin 7
 - v. 1 Green LED on pin 9
 - vi. 2 220ohm resistors.

(This is just an EXAMPLE. Make sure the pins are connected exactly as required!)

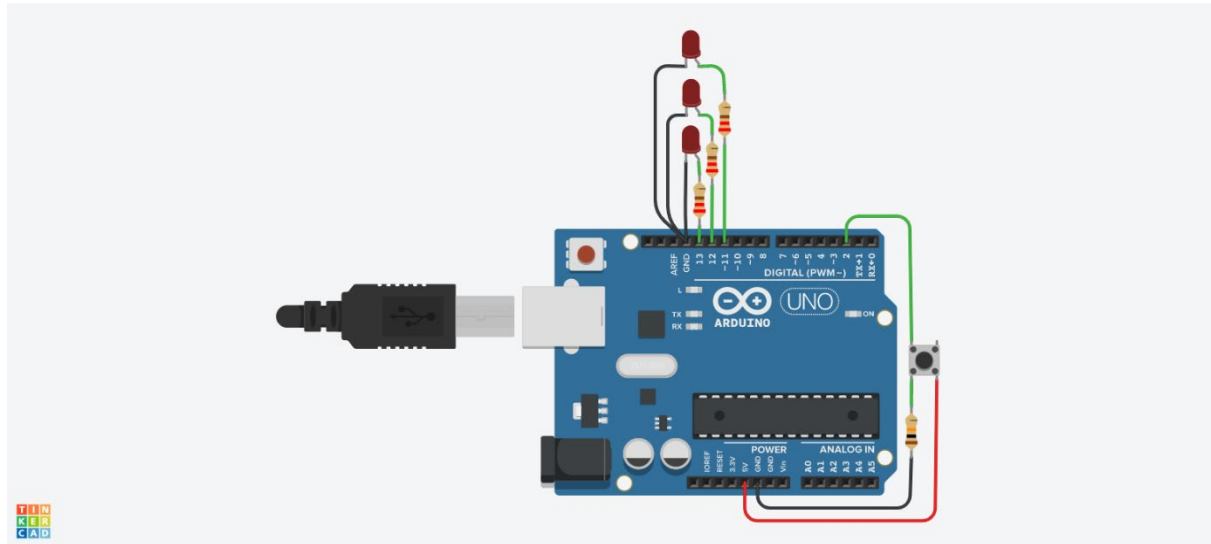


- c. Write a code that does the following:
 - i. If Button #1 (pin 2) is pressed → Green LED ON, otherwise OFF
 - ii. If Button #2 (pin 5) is pressed → Red LED ON, otherwise OFF

5. “5. State Machine”

- a. Before you continue with this exercise – make sure you’ve covered the content of:
“Waiting for Release” (Moodle)!!
- b. Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- c. Construct the following structure:
 - i. Arduino Uno r3.
 - ii. 1 Pushbutton.
 - iii. 1 10Kohm resistor.
 - iv. 3 red LEDs on pins 7, 8, 11 (which represent LED1, LED2, and LED3, respectively).
 - v. 3 220ohm resistors.

(This is just an EXAMPLE. Make sure the pins are connected exactly as required!)



- d. Write a code that simulates a “state machine” defined as the following:

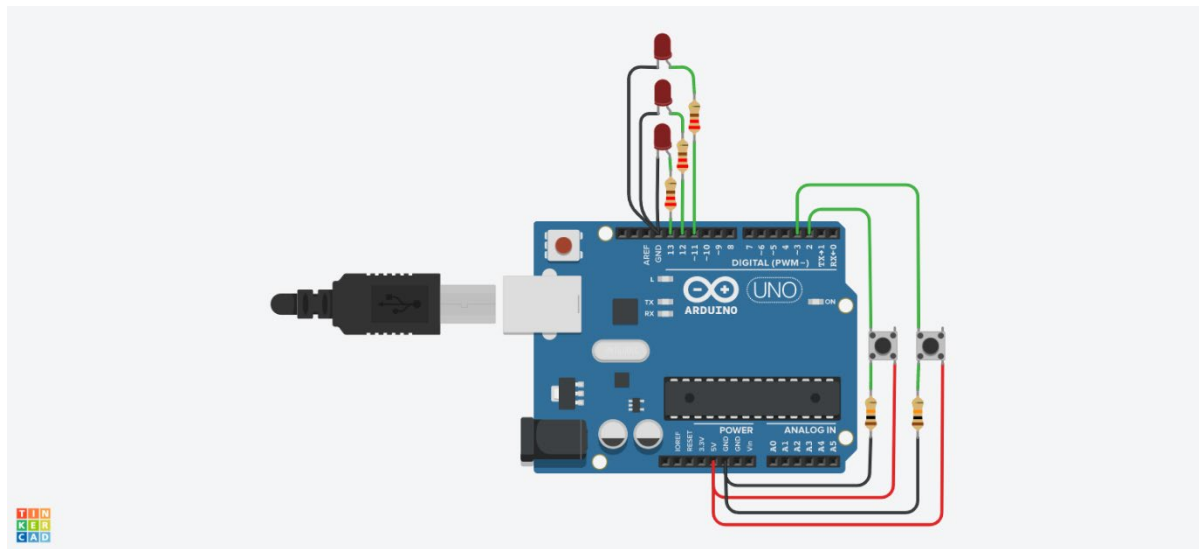
	Button	LED1	LED2	LED3
0	Initial State	0	0	0
1	Press #1	0	0	1
2	Press #2	0	1	0
3	Press #3	0	1	1
4	Press #4	1	0	0
5	Press #5	1	0	1
6	Press #6	1	1	0
7	Press #7	1	1	1

Note: After 7th button press, the state machine should return to the “Initial State”.

6. “6. State Machine (with RESET button)”

- a. Go to “Activities”, “Copy and Tinker” the associated task, and update the file name.
- b. Construct the following structure:
 - i. Arduino Uno r3.
 - ii. 2 Pushbuttons + 2 10Kohm resistors.
 - iii. 3 red LEDs on pins 7, 8, 11
 - iv. 3 220ohm resistors.

(This is just an EXAMPLE. Make sure the pins are connected exactly as required!)



- c. Write a code that simulates a “state machine” defined as the following:

	Button Press / State	LED1	LED2	LED3
0	Initial State	0	0	0
1	Press #1	0	0	1
2	Press #2	0	1	0
3	Press #3	0	1	1
4	Press #4	1	0	0
5	Press #5	1	0	1
6	Press #6	1	1	0
7	Press #7	1	1	1

Note: After 7th button press, the state machine should return to the “Initial State”.

- d. Also, add a “Resetting Functionality” for SW2. When SW2 has been pressed, the state machine should return to the “Initial State”, regardless of the current state. That means when the SW2 is pressed, all the LEDs turn off and the state machine is at its “Initial State” (before Press #1).
(Extra RESET button on pin → return to state 0 and turn all LEDs off)